CROSS-HARMONIC MPI RESTORATION VIA GENERATIVE MODELS

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ABSTRACT

Magnetic Particle Imaging (MPI) offers high-sensitivity, radiation-free biomedical imaging by detecting the nonlinear response of superparamagnetic nanoparticles. However, a fundamental trade-off limits its potential: while higher-order harmonics encode fine spatial details, their rapid signal decay severely compromises the signal-to-noise ratio (SNR), especially at low particle concentrations. We propose a generative restoration framework to dissociate spatial resolution from this intrinsic SNR degradation. Instead of relying on noisy high-harmonic measurements, our approach reconstructs high-fidelity images by learning a mapping from robust, low-harmonic inputs to high-harmonic targets. By employing diffusion and flow-based models, we capture the statistical transformation between these signal domains. Rather than struggling with highly corrupted measurements, our approach capitalizes on a stable low-harmonic scaffold to guide the generative process. This strategy fundamentally breaks the traditional resolution-SNR trade-off, achieving superior structural fidelity and noise robustness in challenging low-concentration regimes compared to conventional solvers.

1. INTRODUCTION

MPI reconstructs the spatial distribution of particles by detecting their nonlinear magnetization response to oscillating magnetic fields [1]. The fundamental imaging mechanism relies on the generation of a field-free point (FFP) or field-free line (FFL), which localizes the particle excitation [2]. Because the particle magnetization follows the Langevin function, the induced voltage signal is rich in higher-order harmonics, which effectively encodes the spatial information required for image formation. However, capturing these essential high-frequency components inherently degrades the signal-to-noise ratio (SNR) due to rapid amplitude decay, a problem that becomes severely restrictive in low-concentration regimes.

Conventional reconstruction techniques fail to adequately overcome this resolution-SNR conflict. Frequency filtering and harmonic truncation reliably suppress noise but inevitably sacrifice spatial resolution, blurring the resulting images. Conversely, regularized inverse reconstructions and physics-based solvers attempt to retain high-harmonic data but often suffer from oversmoothing or severe instability when the underlying signal is dominated by noise [3]. Consequently, achieving sharp, reliable reconstructions from low-concentration measurements remains a persistent challenge.

Rather than forcing a direct inverse solution on noisy data, we formulate this task as an image-to-image translation problem using generative models. Leveraging the paired dataset from the LCR–MPI challenge, our approach completely bypasses the severely degraded high-harmonic measurements during low-concentration inference. Instead, we extract robust, high-SNR low-harmonic components to serve as stable inputs. We then train diffusion and flow-based models to map these blurred observations to high-resolution targets derived from high-concentration, high-harmonic data.

By evaluating a range of generative architectures—including I^2SB [4], RDDM [5], IR-SDE [6], InDi [7], and ResFlow [8] — we demonstrate that learning this cross-condition distribution allows the network to synthesize missing high-frequency structures accurately. This data-driven strategy circumvents the physical limitations of classical MPI reconstruction, offering a robust pathway to high-fidelity imaging under challenging low-concentration conditions.

2. BACKGROUND

Classical MPI image formation relies on deterministic signal processing rather than data-driven priors, offering transparent geometric inversion from raw measurements to spatial images [9]. During acquisition, the receive coil captures a mixture of the tracer's nonlinear response and strong excitation feedthrough [10]. To isolate the relevant particle signature, the time-domain signal is transformed via Fast Fourier Transform (FFT). By filtering out the fundamental driving frequency, the system extracts the higher-order harmonics generated by the Langevin dynamics of the nanoparticles, which carry critical spatial information [11].

In a Field-Free Line (FFL) scanning geometry [12], these harmonic measurements represent a series of line integrals across the imaging field. Collecting these projections over multiple angles mathematically forms a sinogram, equivalent to the Radon transform of the tracer distribution [13]. Therefore, spatial reconstruction typically employs the inverse Radon transform through filtered back-projection (FBP) [14], compensating for low-frequency bias with a ramp filter before back-projecting the data into the image domain [15].

While this analytic framework is highly interpretable, it is notoriously susceptible to noise amplification at higher frequencies [16]. Standard countermeasures such as frequency-domain truncation, apodization windows (e.g., Hann or Hamming), and Tikhonov regularization [17] inherently force a structural compromise. They stabilize the ill-posed inverse problem by suppressing high-frequency artifacts, but simultaneously penalize spatial sharpness, leading to oversmoothed outputs. Thus, in low-concentration scenarios where high-harmonic signals are exceptionally weak, these deterministic solvers fail to balance robust noise suppression with high spatial fidelity [18], [19].

3. METHOD

In our framework, we reconstruct paired data from raw MPI voltage signals acquired under different harmonic and concentration conditions. We define the high-quality target image—reconstructed from high-concentration 5th-harmonic measurements—as the clean initial state x_o. Conversely, the observation derived from low-concentration 3rd-harmonic measurements, which exhibits high SNR but severe spatial blurring, serves as the degraded terminal state x_1.

Rather than learning a naive feed-forward mapping, the generative architectures evaluated in this study mathematically formulate a forward degradation trajectory from the sharp target x_o to the blurred observation x_1. The networks are subsequently trained to invert this corruption process, progressively resolving the high-frequency structural details of x_o by utilizing the stable x_1 measurement as a structural prior. This unified formulation allows us to rigorously evaluate how different generative mechanisms recover fine spatial textures from blurred yet stable inputs.

I^2SB formulates the degradation and restoration as a Schrödinger Bridge, constructing a tractable stochastic process that directly connects the two boundary distributions. Unlike standard diffusion models that degrade x_o into pure Gaussian noise, this approach defines a forward process toward x_1 and learns the corresponding reverse transition. This endpoint conditioning preserves the robust structural information already present in the low-resolution input.

RDDM explicitly decomposes the reverse trajectory into residual and noise diffusion. By treating the transformation from x_1 back to x_0 as a residual regression task, the model focuses entirely on synthesizing the missing high-frequency structural differences, effectively bypassing complex coefficient schedules while leveraging the geometric stability of x_1.

IR-SDE governs the forward degradation from x_0 toward the observation x_1 using a mean-reverting stochastic differential equation. The model is trained to simulate the reverse-time SDE by estimating a time-dependent score function, systematically reconstructing the sharp spatial details of the high-quality target x_0.

InDi constructs a deterministic forward trajectory via a convex interpolation between the clean target x_0 and the degraded input x_1. The reverse restoration is achieved by iteratively predicting x_0 from these intermediate interpolated states, a strategy that mitigates the ill-posedness of the inversion and enhances overall perceptual fidelity.

ResFlow models the degradation as a continuous, deterministic normalizing flow from x_0 to x_1. The restoration is performed by strictly inverting this learned trajectory. By matching a predicted velocity field to the ground-truth vector field along the transformation path, the network establishes a bijective mapping between the high-resolution and low-resolution domains.

4. RESULTS

Method	SSIM	PSNR	NRMSE
I^2SB	**0.6660**	16.8939	0.3794
RDDM	0.6531	18.2035	**0.3360**
IR-SDE	0.6047	16.1892	0.4127
InDi	0.6082	**18.8908**	0.3373
ResFlow	0.6329	16.9683	0.3732

Table 1: Quantitative performance comparison of various reconstruction methods on the MPI Challenge leaderboard.

Table 1 reports the quantitative reconstruction performance of the evaluated generative frameworks on the MPI Challenge leaderboard. The overall success of these methods provides strong empirical evidence that generative architectures can effectively learn complex mappings between entirely distinct physical signal distributions—specifically, across different harmonic bands and tracer concentrations. Among the tested models, I^2SB achieves the highest structural fidelity, highlighted by a leading SSIM of 0.6660. This superior performance aligns directly with the underlying characteristics of our cross-harmonic formulation. Because the low-resolution input already offers a highly reliable spatial draft, the reconstruction task involves exceptionally low structural ambiguity. I^2SB is uniquely — suited to this deterministic regime; by formulating the process as a Schrödinger bridge, it explicitly anchors both the degraded input (x_1) and the high-quality target (x_0). This strong endpoint conditioning restricts the network to minimally perturbed transition dynamics, effectively synthesizing the missing high-frequency details while strictly preserving the robust initial scaffold.

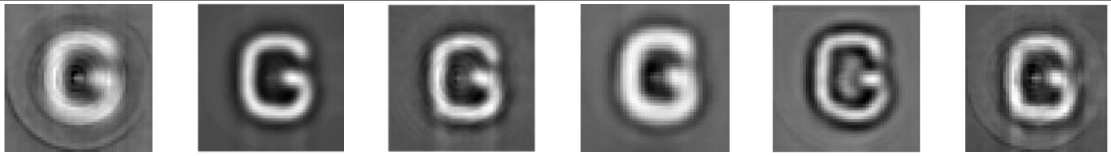
Method	Input(ID#2)	I^2SB	RDDM	IR-SDE	InDi	ResFlow
Visualization Result						

Figure 1: Qualitative reconstruction results for a representative test sample (ID#2). As shown in Figure 1, the I^2SB model effectively removes background noise and artifacts present in the input, producing a cleaner and more accurate representation of the target structure. It is worth noting that all visualizations were generated directly from the model's raw float64 .npy output files to ensure data integrity without loss of precision.

5. ABLATION

In developing the generative restoration pipeline, we initially formulated the task as a direct intra-harmonic denoising problem. By mapping degraded low-concentration 5th-harmonic inputs directly to high-concentration 5th-harmonic targets, this baseline intuitively treats the degradation as a conventional image restoration problem within the same frequency band. This approach inherently avoids the domain shift associated with translating between distinct physical frequency representations. However, empirical evaluations revealed that this strict denoising approach suffers from severe generalization bottlenecks. Under zero-shot conditions involving unseen spatial geometries or altered noise distributions**,** the generative models frequently failed to separate genuine topological features from severe system-induced artifacts. Because the underlying spatial signal in the low-concentration 5th harmonic is almost entirely obfuscated by noise, treating the degradation merely as an accumulation of Gaussian stochastic processes—a standard assumption in diffusion priors—proves theoretically and empirically inadequate for robust recovery.

Recognizing these limitations, we reformulated the objective from ill-posed intra-harmonic denoising to cross-harmonic super-resolution and deblurring. By substituting the severely degraded input with the 3rd-harmonic component, the network leverages a measurement that inherently maintains a high signal-to-noise ratio (SNR). Relying on this robust measurement explicitly bypasses the need to model severe, concentration-dependent noise dynamics. Instead, the 3rd harmonic serves as a reliable geometric scaffold that stably guides the reconstruction process. Even for completely novel particle distributions, the model is no longer forced to synthesize geometry from a noise-dominated observation; rather, it resolves high-frequency details from an already plausible low-frequency draft. As corroborated by our zero-shot visual comparisons (Fig. 2), anchoring the generative process to this stable initial draft fundamentally prevents arbitrary structure generation and guarantees vastly superior generalization across diverse acquisition scenarios.

6. CONCLUSION

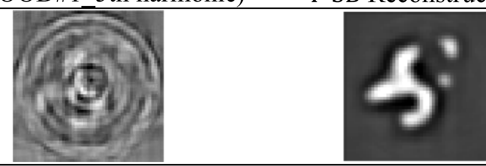
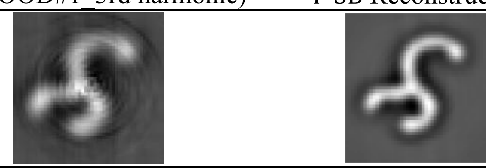
Input(OOD#1_5th harmonic)	I^2SB Reconstruction
	
Input(OOD#1_3rd harmonic)	**I^2SB Reconstruction**
	

Figure 2: Comparison of reconstruction performance between the 3rd and 5th harmonics in **low-concentration MPI**. While the baseline uses the 5th harmonic, our approach utilizes the **3rd harmonic for both training and inference**, resulting in higher resolution and fewer structural distortions.

This study reframes the inherent resolution-noise trade-off in Magnetic Particle Imaging (MPI) as a cross-domain image restoration task. By pairing high-SNR, low-harmonic inputs with high-resolution, high-harmonic targets, we trained generative models to synthesize missing structural details. The fundamental challenge of MPI—that lower, safer tracer concentrations yield overwhelmingly noisy high-frequency signals—renders traditional analytic reconstruction inadequate. By learning the statistical mapping between these distinct signal distributions, diffusion and flow-based models successfully circumvent this physical limitation. Our evaluations confirm that data-driven priors can decouple spatial sharpness from noise amplification, offering robust reconstructions even in challenging scenarios. Ultimately, integrating generative architectures into the MPI pipeline provides a viable pathway toward achieving high-fidelity, low-dose biomedical imaging.

7. LIMITATIONS

A fundamental bottleneck in utilizing diffusion and flow-based architectures for image restoration is their strict requirement for perfectly aligned input-target pairs. While acquiring such supervised data is already notoriously difficult in standard imaging tasks, our proposed framework introduces an even more restrictive constraint. Because our method necessitates mapping across completely different tracer concentrations, constructing a single training pair demands multiple scans under varying physical conditions. Maintaining absolute spatial consistency between a low-concentration input and a high-concentration target drastically increases the logistical complexity of data collection, limiting the immediate practical scalability of this approach.

Beyond the logistical bottlenecks of data acquisition, the theoretical formulation of the employed generative models presents a distinct physical limitation. Standard diffusion-based restoration methods intrinsically frame image degradation as a forward stochastic noising sequence, approaching the reconstruction by learning the reverse denoising trajectory. While mathematically elegant, this generic stochastic approximation cannot perfectly model the true physical noise distribution inherent to actual MPI systems. The real measurement noise is deeply coupled with the nonlinear magnetization dynamics of the nanoparticles and hardware-specific acquisition processes, neither of which are explicitly captured by a purely data-driven prior. By omitting explicit physics-based constraints, such as the system matrix or forward measurement operators, the network's ability to guarantee absolute structural exactness is fundamentally bounded. This lack of precise physical modeling ultimately restricts the overall reconstruction fidelity, as evidenced by our SSIM scores plateauing near 0.65, demonstrating the inherent difficulty of fully synthesizing high-frequency details relying on stochastic approximations alone.

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